

Supplementary Material for “Understanding Q-Learning and Deep Q-Learning in 2026: A Methodological and Empirical Survey”

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This document collects the auditable artifacts and extended material referenced by the main paper: the systematic search log (supporting §III), the full per-method method-type index (§III, §IV, §VIII), the complete per-game Atari benchmark tables (§V), the full repository coverage matrix (§VII), notation and selected derivations (§IV.I), and the full configuration of the tabular experiment (§VI). Section and table references of the form “§X” point to the main manuscript.

S1. SYSTEMATIC SEARCH LOG AND PRIOR-ART SWEEP

This appendix documents the systematic literature search supporting the methodology of §III. It records the databases queried, the date range, the search strings, the inclusion and exclusion outcome, and a structured comparison against recent reinforcement-learning and Q-learning surveys. It is provided so that the screening flow and the survey-positioning claims in §III can be independently audited.

S1.1 Databases and Date Range

Three databases were queried. Google Scholar provided broad coverage and citation-graph traversal; arXiv (categories cs.LG and cs.AI) served as the primary preprint source; and Semantic Scholar supported citation-relationship analysis, including forward and backward citation expansion seeded from the closest-matching surveys.

The recent-work sweep covered December 2025 through May 2026, with a cutoff of 2026-05-15. A recency filter of 2018 onward was applied to deep-RL methods; foundational tabular and theoretical results were admitted with no date floor.

S1.2 Search Strings

Queries combined topical anchors with axis-specific terms. The topical anchors were “Q-learning”, “deep Q-network”, and “DQN”. These were conjoined with axis-specific terms including “overestimation bias”, “prioritized replay”, “exploration bonus”, “distributional reinforcement learning”, “offline reinforcement learning”, “value decomposition”, “distributed reinforcement learning”, “meta-reinforcement learning”, and “function approximation stability”. A complementary set of queries targeted recent survey literature directly, combining

“reinforcement learning” and “Q-learning” with “survey”, “overview”, and “taxonomy” to identify works whose scope could overlap that of this paper.

S1.3 Inclusion and Exclusion Outcome

Candidates were screened against the five inclusion criteria stated in §III: method centrality, mechanism distinctness, empirical validation, cross-reference impact, and reproducibility. Initial identification produced approximately 200 records; title and abstract screening narrowed this to approximately 120; full-text screening against the five criteria yielded approximately 80 works receiving per-method treatment, plus approximately 20 cited at family or cross-reference level. Excluded records comprised application papers using Q-learning as a black box, workshop-only or non-archival preprints without follow-up, and methods superseded by equivalent-effort alternatives.

A dedicated survey-comparison sweep was run over the December 2025–May 2026 window to confirm the positioning claims of §III. No survey published in the 2024–2026 window adopts a multi-axis problem-first organization of the Q-learning family unifying tabular and deep methods. The nearest organizational precedent applies a problem-first structure along a single axis (distribution shift) and is restricted to offline RL. The works identified by this sweep are summarized below.

S1.4 Prior Surveys Identified

Four works were identified as overlapping or adjacent in scope. Three are surveys; one is an empirical implementation study rather than a survey and is noted as such. The following table summarizes each work’s scope, its overlap with the present survey, and the points of differentiation.

Additional works surfaced by the sweep were assessed and excluded as non-competing on scope grounds: a taxonomy of RL for robotics and control (continuous control, minimal DQN coverage), a survey of explainable deep RL (organized by explanation level), a study of Q-learning for metaheuristic optimization (Q-learning as a controller rather than core theory), and several application-domain surveys (network intrusion detection, real-world robotics successes, reward models). None adopts a problem-first Q-learning taxonomy and none overlaps the contributions of this survey.

Work	Scope	Overlap with this survey	Differentiation
Ghasemi, Moosavi and Ebrahimi 2024/2025 [@ghasemi_2024_rlsurvey]	Broad RL across tabular and deep methods, organized by method family (value-based, policy, actor-critic), with practical challenges as a secondary lens	Covers core Q-learning methods (Q-learning, Double Q-learning, DQN, DDQN, Dueling DQN) and unifies tabular and deep treatment	Organized by method family rather than problem axis; finer method-type categories (prioritized replay, distributional, Rainbow) and the repository and Atari analyses of this survey are not present
Springer NCAA 2026 distribution-shift survey [@springer_2026_distshift]	Offline RL only, organized problem-first along a single axis (distribution shift), with four method buckets (Q-value restriction, uncertainty-based Q-restriction, policy constraint, uncertainty-based policy constraint)	Shares a problem-first organizing principle and addresses the distribution-shift axis treated in §IV.E	Single-axis and restricted to offline RL; this survey generalizes the problem-first structure to multiple axes spanning tabular and deep Q-learning
Murphy 2024/2025 [@murphy_2024_rloverview]	Textbook-style overview covering value-based, policy, model-based, multi-agent, and LLM-plus-RL methods, organized by method type	General coverage of value-based methods, including an explicit LLM-RL angle that intersects §IV.J	Organized by method type; does not provide a Q-learning-specific DQN taxonomy, repository comparison, or extracted Atari benchmarks
Hundal, Xiao, Cao, Dong and Rigger 2025 [@hundal_2025_interchangeable]	Empirical benchmark of PPO across Stable-Baselines3, CleanRL, Baselines, RLlib, and Tianshou over 56 Atari games (implementation audit, not a survey)	Studies the same open-source repository landscape examined in the repository analysis of §VII	An empirical reproducibility audit of one algorithm (PPO); this survey’s repository treatment is taxonomic, mapping algorithmic coverage across repositories to identify Q-learning implementation gaps. The two approaches are complementary

S2. FULL METHOD-TYPE INDEX

This supplement is the full reference index promised by §III, §IV, and §VIII. It supports readers who approach the material through the conventional method-type taxonomy used in prior Q-learning surveys [1–4]. The eight-axis problem-first organization of §IV is the analytical spine of the paper; the six method-type categories are preserved as a secondary indexing system. A reader interested in all distributional methods navigates via the method-type view (the Statistical Methods rows below); a reader interested in all methods addressing brittle exploration navigates via §IV.C. The two views are complementary.

S2.1. Full method-to-axis mapping

For each method covered in §IV, the table gives its **Method type** (the conventional grouping: Statistical Methods, Q-Function Computation, Memory/Replay, Ensemble-Based, Model-Based, Pure Q-Learning), its **Primary axis** (the weakness W_i from §II.B that the method principally addresses, with its full-treatment section), and **Secondary axes** (weaknesses it incidentally mitigates).

S2.2. Methods outside the six method-type categories

Eighteen methods covered in §IV have no row in the conventional six-category taxonomy and do not appear in the main paper’s row-spine tables. They are treated in §IV under the new axes; the table below maps them to the method-type taxonomy as they *would* have been classified if the conventional categories were extended to admit them.

The “New category” entries cannot be cleanly placed in the six method-type categories: they are Q-learning families that emerged after the taxonomy was conventionalized. Their absence is a central argument in §I for the structural pivot — the six categories were defined before these families existed,

and accommodating them requires adding categories that were never load-bearing. The eight-axis structure has principled homes for all eighteen methods without category-set expansion.

S2.3. Reverse view — method type to §IV section coverage

The table inverts S2.1: for each method-type category, what fraction of its constituent methods land in each axis-section of §IV.

Three patterns emerge from the inverted view:

1. *Memory/Replay* is the most internally coherent method-type category: all four methods land in §IV.B (sample inefficiency). The method-type category and the axis category coincide.
2. *Ensemble-Based* fragments: ensembles serve overestimation control (§IV.A) and exploration (§IV.C) via the same architectural mechanism but different mechanisms-of-use. The axis view distinguishes them; the method-type view does not.
3. *Statistical Methods* fragments into noise-based exploration (§IV.C) and distributional credit-assignment (§IV.D) — the sharpest single demonstration of why the conventional method-type taxonomy conflates mechanism with effect.

These patterns are the data backing the problem-first organization, verifiable method-by-method.

S3. FULL ATARI PER-GAME BENCHMARK TABLES

The two tables below give the full per-game Atari scores that main §V defers to this supplement. All values are scores exactly as reported in the original papers introducing each method; we have not re-run any experiment or rescaled any number. Evaluation protocols differ across publications — no-op starts versus sticky actions, frameskip settings, training-frame budgets, and seed counts all vary — so the cells are

Method (year)	Method type	Primary axis	Secondary axes
Parameter Space Noise (2017)	Statistical	§IV.C (W3)	—
NoisyNet (2018)	Statistical	§IV.C (W3)	—
C51 (2017)	Statistical	§IV.D (W4)	§IV.A
QR-DQN (2018)	Statistical	§IV.D (W4)	§IV.A
IQN (2018)	Statistical	§IV.D (W4)	§IV.A
FQF (2019)	Statistical	§IV.D (W4)	§IV.A
Double Q-Learning (2010)	Q-Function Comp.	§IV.A (W1)	§V foundations
Nature DQN (2015)	Q-Function Comp.	§IV.H (W8)	§V foundations
Deep Recurrent Q (DRQN, 2015)	Q-Function Comp.	§IV.G (W7)	—
Double DQN (2016)	Q-Function Comp.	§IV.A (W1)	—
Dueling DQN (2016)	Q-Function Comp.	§IV.H (W8)	§IV.A (incidental)
Rainbow (2018)	Q-Function Comp.	§IV.B (W2)	§IV.A, §IV.C, §IV.D, §IV.H
MCTS for FrozenLake (2024)	Q-Function Comp.	§V planning baseline	—
Cognitive Belief-Driven Q (2025)	Q-Function Comp.	§IV.C (W3)	§IV.A
DQN (2013)	Memory/Replay	§IV.B (W2)	§V foundations
Prioritized ER (PER, 2016)	Memory/Replay	§IV.B (W2)	—
DQfD (2018)	Memory/Replay	§IV.B (W2)	§IV.C (demos for exploration)
Memory-Efficient DQN (MeDQN, 2023)	Memory/Replay	§IV.B (W2)	§IV.H (consolidation)
Bootstrapped DQN (2016)	Ensemble-Based	§IV.C (W3)	§IV.A
Ensemble Bootstrapped Q (EBQL, 2021)	Ensemble-Based	§IV.A (W1)	§IV.C
UCB Q-Ensemble (2018)	Ensemble-Based	§IV.C (W3)	§IV.A
Value/Policy Iteration (1960s)	Model-Based	§V foundations	—
Bayesian Q-Learning (1998)	Model-Based	§V foundations	§IV.C (Thompson sampling)
Expected SARSA (2009)	Model-Based	§V foundations	—
Posterior Sampling DQN (2023)	Model-Based	§IV.C (W3)	§IV.A
Q-Learning (Watkins 1992)	Pure Q-Learning	§V foundations	—
SARSA (1994)	Pure Q-Learning	§V foundations	—
Multi-Step Q-Learning (1996)	Pure Q-Learning	§IV.D (W4)	§V foundations
Neural Fitted Q (NFQ, 2005)	Pure Q-Learning	§V foundations	§IV.B (batch updates)
Parallel Q Learning (PQN, 2024)	Pure Q-Learning	§IV.H (W8)	§IV.G (parallelism)

not mutually calibrated. These tables must therefore be read with the reliable-style caveats stated in main §V [5]: as a record of what each paper reported under its own protocol, not as a leaderboard, and never as a per-cell “best method” ranking.

Per-game scores for the Reaction-Time Control, Strategic Planning, Sparse Rewards, and Dense Rewards task categories, grouped by method type.

Per-game scores for the Large Observation Space, Partially Observable, and Stochastic Environments task categories, grouped by method type.

S4. FULL REPOSITORY COVERAGE MATRIX

This is the full method-level coverage matrix that main §VII defers to supplementary material, spanning six widely-used deep RL repositories — Tianshou [6], XuanCe [7], CleanRL [8], DQN Zoo [9], Stable Baselines3 [10], and RLlib [11]. It is a dated snapshot reflecting the state of each repository as of 2026-05-01, for the audited version ranges Tianshou 0.5.x, XuanCe 1.x, CleanRL master (commit-pinned during the audit), DQN Zoo (current head), Stable Baselines3 2.x, and RLlib 2.x. Support is marked by named-algorithm presence — a method counts as supported only if the repository implements it as a named algorithm with appropriate hyperparameter exposure — so the named-vs-feature-equivalent caveat from main §VII applies, and feature-equivalent availability of compound methods is consistently higher than the matrix indicates.

The design trade-offs behind these coverage patterns are summarized below; this per-repository table is more detailed than the prose summary carried in main §VII. The taxonomic-

coverage view here is complementary to the empirical-reproducibility audit of Hundal et al. [12], which compares a single algorithm across repositories under nominally equivalent configurations.

These priorities are not in conflict but occupy different positions on a Pareto frontier: a repository optimizing for one property necessarily relaxes another. Selection of a repository is therefore itself an axis decision — Tianshou serves overestimation-method exploration well, RLlib serves at-scale deployment, and CleanRL serves educational demonstration — and no single repository currently serves the Q-learning research community comprehensively.

S5. NOTATION AND SELECTED DERIVATIONS

This supplement consolidates the symbol conventions used throughout the paper and supplies the derivations that the main text references but does not work through. The intent is to keep the §IV bodies focused on mechanism and trade-off while still grounding the claims for readers who want the underlying math. It is the formal companion promised by §IV.I and the §I reading guide.

S5.1. Notation

The symbols below appear throughout the paper. Section-local deviations from these conventions are flagged at first use.

Boldface lower-case (e.g., **a**, **s**) denotes joint quantities in multi-agent settings (§IV.F).

Method (year)	Would-be method type	Primary axis
Maximin Q-Learning (2020)	Q-Function Comp.	§IV.A (W1)
REDQ (2021)	Ensemble-Based	§IV.A (W1)
HER (2017)	Memory/Replay	§IV.B (W2)
RND (2018)	Statistical (intrinsic motivation)	§IV.C (W3)
Go-Explore (2019/2021)	Memory/Replay (archive)	§IV.C (W3)
BCQ (2019)	New category — Offline RL	§IV.E (W5)
BRAC (2019)	New category — Offline RL	§IV.E (W5)
AWAC (2020)	New category — Offline RL	§IV.E (W5)
CQL (2020)	New category — Offline RL	§IV.E (W5)
IQL (2021)	New category — Offline RL	§IV.E (W5)
EDAC (2021)	New category — Offline RL + Ensemble	§IV.E (W5)
VDN (2018)	New category — Multi-Agent	§IV.F (W6)
QMIX (2018)	New category — Multi-Agent	§IV.F (W6)
QPLEX (2020)	New category — Multi-Agent	§IV.F (W6)
QTRAN (2019)	New category — Multi-Agent	§IV.F (W6)
Ape-X (2018)	New category — Distributed	§IV.G (W7)
R2D2 (2019)	New category — Distributed + Recurrent	§IV.G (W7)
Agent57 (2020)	New category — Distributed + Meta-policy	§IV.G (W7)
MAML-Q (2017/2019)	New category — Meta-RL	§IV.G (W7)
Munchausen DQN (2020)	Q-Function Comp. (would have been)	§IV.H (W8)

Method type	Primary axis distribution
Statistical Methods (6 methods)	2 → §IV.C (Param Noise, NoisyNet); 4 → §IV.D (C51, QR-DQN, IQN, FQF)
Q-Function Computation (8 methods)	2 → §IV.A (Double Q, Double DQN); 2 → §IV.H (Nature DQN, Dueling); 1 → §IV.B (Rainbow); 1 → §IV.C (CBDQ); 1 → §IV.G (DRQN); 1 → §V (MCTS)
Memory/Replay (4 methods)	4 → §IV.B (DQN, PER, DQfD, MeDQN)
Ensemble-Based (3 methods)	1 → §IV.A (EBQL); 2 → §IV.C (Bootstrapped DQN, UCB Q-Ensemble)
Model-Based (4 methods)	1 → §IV.C (PSDQN); 3 → §V (VI/PI/MPI, Bayesian Q, Expected SARSA)
Pure Q-Learning (5 methods)	4 → §V (Q-learning, SARSA, NFQ, multi-step); 1 → §IV.H (PQN)

S5.2. Maximum-of-noisy-estimators bound (referenced by §IV.A)

Section IV.A invokes the result that the maximum of noisy estimators is biased upward as the source of Q-learning’s overestimation. [13] formalize this for the related setting of decision analysis. The relevant inequality for our purposes:

Let $\hat{Q}_a = Q_a^* + \xi_a$ for $a \in \mathcal{A}$, where $\{\xi_a\}$ are zero-mean noise variables with positive variance. Then

$$\mathbb{E} \left[\max_{a \in \mathcal{A}} \hat{Q}_a \right] \geq \max_{a \in \mathcal{A}} Q_a^*,$$

with strict inequality when $|\mathcal{A}| \geq 2$ and at least two ξ_a have positive variance.

Proof sketch. For each fixed realization $\{\xi_a\}$, the function $\xi \mapsto \max_a (Q_a^* + \xi_a)$ is convex in ξ . Jensen’s inequality applied to the expectation over ξ yields

$$\mathbb{E}_\xi \left[\max_a (Q_a^* + \xi_a) \right] \geq \max_a \mathbb{E}_\xi [Q_a^* + \xi_a] = \max_a Q_a^*.$$

Strict inequality follows from non-degeneracy of the maximum under random perturbation. Iterated through the Bellman backup, the bias compounds, motivating the decoupling methods of §IV.A.B.1 and the ensemble methods of §IV.A.B.2.

S5.3. Categorical distributional projection (referenced by §IV.D)

C51 [14] maintains a return distribution as a categorical distribution over $N = 51$ fixed atoms in $[V_{\min}, V_{\max}]$. After applying the distributional Bellman update $\hat{\mathcal{T}}Z(s, a) = r + \gamma Z(s', a^*)$, the resulting distribution generally has support outside the fixed atom set; the projection operator Φ maps it back.

Definition. Let $\{z_i\}_{i=0}^{N-1}$ be the atom locations with $z_i = V_{\min} + i\Delta z$, $\Delta z = (V_{\max} - V_{\min})/(N - 1)$. Given a candidate distribution with probability mass p_j at locations $\tilde{z}_j$, the projection Φ produces a distribution on $\{z_i\}$ with probabilities

$$(\Phi(\tilde{z}, p))_i = \sum_j p_j \cdot \max \left(0, 1 - \frac{|\tilde{z}_j - z_i|}{\Delta z} \right),$$

with $\tilde{z}_j$ clipped to $[V_{\min}, V_{\max}]$ before projection. Probability mass is conserved by construction (the weights of any $\tilde{z}_j$ sum to 1 across the two adjacent atoms).

The KL loss in C51’s training objective is then

$$\mathcal{L}_{C51} = \text{KL}(\Phi(\hat{\mathcal{T}}Z_{\bar{\theta}}) \parallel Z_{\theta}),$$

where Z_{θ} is the predicted distribution under the online parameters and $\bar{\theta}$ are the target parameters. Note that the projection step is *non-contractive* in the KL norm — the contraction results of §IV.I.C.1 hold in the Wasserstein metric instead.

S5.4. Wasserstein contraction of the distributional Bellman operator (referenced by §IV.I.C.1)

[14] establish that the distributional Bellman operator $\mathcal{T}_{\pi}^d$ is a γ -contraction in the maximal form of the Wasserstein- p distance over return distributions.

Statement. For two return distribution functions $Z_1, Z_2 : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{P}(\mathbb{R})$,

$$\bar{W}_p(\mathcal{T}_{\pi}^d Z_1, \mathcal{T}_{\pi}^d Z_2) \leq \gamma \bar{W}_p(Z_1, Z_2),$$

where $\bar{W}_p(Z_1, Z_2) = \sup_{s,a} W_p(Z_1(s, a), Z_2(s, a))$ and W_p is the Wasserstein- p distance between scalar distributions.

TABLE I
REPORTED ATARI BENCHMARK PERFORMANCE (RAW PER-GAME SCORES), PART I — REACTION-TIME CONTROL / STRATEGIC PLANNING / SPARSE REWARDS / DENSE REWARDS. “—” INDICATES THE ORIGINAL PAPER DID NOT REPORT A SCORE FOR THAT GAME.

Method (year)	Method type	Breakout	Sp. Inv.	Ms. Pac-Man	Q*bert	Montezuma	Pitfall!	Boxing	Enduro
Param Space Noise (2017)	Statistical	390	1,205	—	7,525	0	-100	—	1,672
C51 (2017)	Statistical	748	5,747	3,415	23,784	0	0	98	3,454
NoisyNet (2018)	Statistical	516	2,186	2,722	15,545	3	0	89	1,240
QR-DQN (2018)	Statistical	742	20,972	5,821	572,510	0	0	100	2,355
IQN (2018)	Statistical	734	28,888	6,349	25,750	0	0	100	2,359
FQF (2019)	Statistical	854	140	7,632	27,524	0	0	98	2,371
Nature DQN (2015)	Q-Func. Comp.	401	1,976	2,311	10,596	0	—	72	302
Deep Recurrent Q (2015)	Q-Func. Comp.	—	—	2,048	—	—	—	—	—
Double DQN (2016)	Q-Func. Comp.	375	3,155	3,210	14,875	0	—	82	320
Dueling DQN (2016)	Q-Func. Comp.	345	6,427	6,284	19,220	0	0	99	2,258
Rainbow DQN (2018)	Q-Func. Comp.	418	18,789	5,380	33,818	384	0	100	2,126
CBDQ (2025)	Q-Func. Comp.	—	—	—	—	—	—	—	—
DQN (2013)	Memory/Replay	168	581	—	1,952	—	—	—	470
Prioritized ER (2016)	Memory/Replay	481	1,697	965	12,741	44	-194	70	1,266
DQfD (2018)	Memory/Replay	308	—	4,696	21,793	4,638	57	99	2,200
MeDQN (2023)	Memory/Replay	—	—	—	—	—	—	—	—
Bootstrapped DQN (2016)	Ensemble	855	2,893	2,983	15,093	100	—	93	1,591
UCB Q-Ensemble (2018)	Ensemble	411	2,627	3,425	14,198	4	-1	98	2,753
Ensemble Bootstrapping (2021)	Ensemble	406	—	—	14,384	—	—	—	—
Posterior Sampling DQN (2023)	Model-Based	46	511	1,824	4,245	0	-44	79	363
Parallel Q (PQN, 2024)	Pure Q	515	18,451	5,568	31,717	0	-89	100	2,349

Proof sketch. The distributional Bellman operator decomposes into three steps: (i) sample a transition (s, a, r, s') from P ; (ii) shift and scale the return distribution at the next state by the affine map $z \mapsto r + \gamma z$; (iii) integrate over the policy $\pi(a' | s')$.

Wasserstein- p is preserved under expectation (step iii) and satisfies $W_p(\delta_r + \gamma X, \delta_r + \gamma Y) = \gamma W_p(X, Y)$ under affine transformations (step ii). Taking the supremum over (s, a) on both sides yields the stated contraction.

The KL divergence used by C51’s loss does *not* satisfy an analogous contraction; this is the technical motivation for QR-DQN’s switch to quantile-regression losses, where the

Wasserstein- ∞ contraction is preserved [15].

S5.5. Pessimism lower-bound argument for offline RL (referenced by §IV.E and §IV.I.C.3)

Pessimistic value iteration constructs a lower confidence bound (LCB) on Q-values at each Bellman backup. The canonical version due to [16] runs

$$\hat{Q}^{k+1}(s, a) = r(s, a) + \gamma \sum_{s'} \hat{P}(s' | s, a) \max_{a'} \hat{Q}^k(s', a') - b(s, a),$$

TABLE II
REPORTED ATARI BENCHMARK PERFORMANCE, PART II — LARGE OBSERVATION SPACE / PARTIALLY OBSERVABLE / STOCHASTIC ENVIRONMENTS.

Method (year)	Method type	River Raid	Priv. Eye	Frostbite	Hero	Zaxxon	Berzerk
Param Space Noise (2017)	Statistical	—	100	1,310	—	8,050	—
C51 (2017)	Statistical	17,322	15,095	3,965	38,874	10,513	1,645
NoisyNet (2018)	Statistical	9,425	3,712	753	6,246	6,920	905
QR-DQN (2018)	Statistical	17,571	350	4,384	21,395	13,112	3,117
IQN (2018)	Statistical	17,765	200	4,324	28,386	21,772	1,053
FQF (2019)	Statistical	23,561	140	16,473	30,926	15,180	12,422
Nature DQN (2015)	Q-Func. Comp.	8,316	1,788	328	19,950	4,977	—
Double DQN (2016)	Q-Func. Comp.	12,015	670	242	20,357	10,182	—
Dueling DQN (2016)	Q-Func. Comp.	21,163	103	4,673	20,818	13,886	3,409
Rainbow DQN (2018)	Q-Func. Comp.	—	4,234	9,591	55,887	22,210	2,546
Prioritized ER (2016)	Memory/Replay	10,206	2,202	289	15,151	9,501	644
DQfD (2018)	Memory/Replay	18,735	42,457	—	105,929	—	—
Bootstrapped DQN (2016)	Ensemble	12,845	1,813	2,181	21,021	11,492	—
UCB	Ensemble	15,622	1,252	1,903	—	3,695	—
Q-Ensemble (2018)	Ensemble	—	100	—	—	—	—
Ensemble Bootstrapping (2021)	Ensemble	—	100	—	—	—	—
Posterior Sampling DQN (2023)	Model-Based	3,858	68	929	7,965	4,413	386
Parallel Q (PQN, 2024)	Pure Q	28,764	100	7,314	26,099	23,538	18,542

where $b(s, a)$ is a *bonus* term inversely proportional to the visitation count $n(s, a)$ in the offline dataset. Concretely, $b(s, a) = c\sqrt{1/n(s, a)}$ for a problem-dependent constant c .

Suboptimality bound. Let $\hat{\pi}$ be the policy greedy with respect to the fixed point $\hat{Q}^*$ of the LCB iteration, and let π be any policy that the offline dataset supports. Then

$$V^\pi(s_0) - V^{\hat{\pi}}(s_0) \leq \tilde{O}\left(\sqrt{\frac{1}{n}}\right),$$

where n is the total dataset size, s_0 is the initial state, and the $\tilde{O}$ hides logarithmic factors and the problem horizon.

The key insight. Pessimism makes $\hat{Q}^*(s, a)$ a *lower bound* on V^π for any supported policy. The greedy policy with respect to $\hat{Q}^*$ therefore competes with the best *supported* policy, not with the unrestricted optimum. Crucially, this bound holds *without coverage assumptions* on the behavior policy — it adapts gracefully when the dataset covers only a small subset of the state space.

CQL’s [17] conservative penalty is a tractable relaxation of this LCB construction: rather than maintaining explicit confidence bounds, CQL adds a regularizer that drives $\hat{Q}$ down at OOD actions, achieving the same pessimism guarantee under specific assumptions on the regularization weight α . IQL’s expectile regression achieves a related effect via the asymmetric quantile loss without explicit OOD-action sampling.

S5.6. QPLEX IGM completeness (referenced by §IV.F and §IV.I.C.5)

The IGM (Individual-Global-Max) constraint requires that the joint argmax of Q_{tot} coincide with the per-agent argmaxes of $\{Q_i\}$:

$$\arg \max_{\mathbf{a}} Q_{\text{tot}}(\mathbf{s}, \mathbf{a}) = \left(\arg \max_{a_1} Q_1, \dots, \arg \max_{a_N} Q_N \right).$$

QMIX’s sufficient condition. Monotonic mixing — $\partial Q_{\text{tot}} / \partial Q_i \geq 0$ for all i — implies IGM but is not necessary. There exist IGM-compatible Q_{tot} that QMIX cannot represent.

QPlex’s claim [18]. Decomposing $Q_{\text{tot}} = \sum_i V_i(\tau_i) + A_{\text{tot}}(\mathbf{s}, \mathbf{a})$ with A_{tot} produced by a duplex-dueling mixer admits *every* IGM-compatible Q_{tot} . The proof constructs an explicit duplex-dueling representation for any IGM-satisfying joint Q_{tot} by reducing to per-agent advantage representations that satisfy IGM individually.

Significance. The result identifies QPLEX’s representational capacity as the ceiling of value-decomposition methods. QTRAN’s auxiliary-loss approach reaches the same ceiling via a different construction but at the cost of training instability discussed in §IV.F.B.4. The representation–training-dynamics gap that QTRAN exposes is precisely the open question listed in §IV.I.E item 4.

TABLE III
REPOSITORY SUPPORT FOR DEEP Q-LEARNING ALGORITHMS, GROUPED BY METHOD-TYPE. ● = IMPLEMENTED; ○ = NOT IMPLEMENTED.

Method-type	Method (year)	Tianshou	XuanCe	CleanRL	SB3	RLlib	DQN Zoo
Statistical	Param Space Noise (2017)	○	●	○	○	○	○
Statistical	C51 (2017)	●	●	●	○	●	●
Statistical	NoisyNet (2018)	○	○	○	○	●	○
Statistical	QR-DQN (2018)	●	●	○	●	○	●
Statistical	IQN (2018)	●	○	○	○	○	●
Statistical	FQF (2019)	●	○	○	○	○	○
Q-Func. Comp.	Nature DQN (2015)	●	●	●	●	●	●
Q-Func. Comp.	DRQN (2015)	○	○	○	○	○	○
Q-Func. Comp.	Double DQN (2016)	●	●	○	○	●	●
Q-Func. Comp.	Dueling DQN (2016)	●	●	○	●	●	○
Q-Func. Comp.	Rainbow DQN (2018)	●	○	○	○	●	●
Q-Func. Comp.	CBDQ (2025)	○	○	○	○	○	○
Memory/Replay	DQN (2013)	○	○	○	○	○	○
Memory/Replay	Prioritized ER (2016)	○	●	○	○	●	●
Memory/Replay	DQfD (2018)	○	○	○	○	○	○
Memory/Replay	McDQN (2023)	○	○	○	○	○	○
Ensemble	Bootstrapped DQN (2016)	○	○	○	○	○	○
Ensemble	UCB Q-Ensemble (2018)	○	○	○	○	○	○
Ensemble	Ensemble Bootstrapping (2021)	○	○	○	○	○	○
Model-Based	Posterior Sampling DQN (2023)	○	○	○	○	○	○
Pure Q	Parallel Q (PQN, 2024)	○	○	○	○	○	○

S5.7. Tabular Q-learning convergence (referenced by §V and §IV.I.B.1)

The canonical Watkins & Dayan (1992) convergence proof for tabular Q-learning runs as follows.

Setting. A finite MDP $\langle S, \mathcal{A}, P, R, \gamma \rangle$ with bounded rewards. The Q-learning update at step t is

$$Q_{t+1}(s, a) = (1 - \alpha_t(s, a))Q_t(s, a) + \alpha_t(s, a)(r_t + \gamma \max_{a'} Q_t(s, a') - Q_t(s, a)).$$

Convergence conditions.

1. **Coverage:** Every $(s, a) \in S \times \mathcal{A}$ is visited infinitely often.
2. **Learning rate decay:** $\sum_t \alpha_t(s, a) = \infty$ and $\sum_t \alpha_t(s, a)^2 < \infty$ for every (s, a) .
3. **Bounded rewards:** $|r_t| \leq R_{\max}$ for all t .

Theorem. Under conditions 1–3, $Q_t \rightarrow Q^*$ with probability 1.

Proof outline. Define the Bellman optimality operator $T(s, a) = R(s, a) + \gamma \mathbb{E}_{s'}[\max_{a'} Q(s', a')]$. The Q-

TABLE IV
COMPARISON OF POPULAR DEEP RL REPOSITORIES — DESIGN PROS AND CONS.

Repository	Pros	Cons
Tianshou	Dual API (high-level and low-level); emphasis on reproducibility with agent-level tests; native TensorBoard support	Logs/results not available for all algorithms; reproducibility guarantees not empirically verified across full benchmark suite
XuanCe	Modular YAML config files; W&B integration for hyperparameter tuning; high modularity for code reuse	Default hyperparameters often diverge from original papers; incomplete support across environments; steep learning curve
CleanRL	Single-file implementations; easy to audit and understand; lightweight and minimal dependencies	Limited support for large-scale experiments; less modular, harder to extend
Stable Baselines3	Clean API with sklearn-style interface; well-maintained and widely adopted; compatible with VecEnv, Gym, etc.	Focused more on policy-gradient methods; limited Q-learning variants beyond DQN
RLlib	Distributed training out-of-the-box; Ray Tune integration; production-grade scalability	High complexity and heavyweight; harder to debug or customize individual components
DQN Zoo	Faithful reimplementation of DQN variants; reproducibility aligned with DeepMind practices; highly organized training logs and configs	Focused exclusively on DQN family; less beginner-friendly

learning update can be rewritten as a stochastic approximation to the fixed-point iteration $Q \leftarrow \mathcal{T}^*Q$. Since $\mathcal{T}^*$ is a γ -contraction in the ℓ_∞ norm and has unique fixed point Q^* , classical stochastic approximation theory [Robbins-Monro 1951; Tsitsiklis 1994] yields almost-sure convergence under conditions 1–3.

Limitations. The proof requires *tabular* representation: each $Q(s, a)$ is a separately-updated scalar, and the contraction argument uses the unique-fixed-point property of $\mathcal{T}^*$. Function approximation breaks both requirements — updates at one (s, a) now affect others, and the Bellman operator composed with function-approximation projection is not in general a contraction. This is the foundational source of the deadly-triad instability of §IV.H.A and §IV.I.B.2.

S5.8. Pointers for further reading

The derivations above are the ones most directly load-bearing for the claims in §IV. Readers seeking deeper treatment are referred to the following canonical sources:

Symbol	Meaning
s, a	State and action at the current time step
s', a'	Successor state and action
r	Immediate reward
$\gamma \in [0, 1)$	Discount factor
$\mathcal{S}, \mathcal{A}$	State and action spaces
$P(s' s, a)$	Transition probability function
$R(s, a)$	Reward function (expected immediate reward)
π	Policy; $\pi(a s)$ for stochastic, $\pi(s)$ for deterministic
V^π, Q^π	State-value and action-value functions under π
V^*, Q^*	Optimal value and action-value functions
$\mathcal{T}, \mathcal{T}^*$	Bellman and Bellman-optimality operators
θ	Online network parameters
θ^-	Target network parameters
α	Learning rate
ϵ	ϵ -greedy exploration parameter; or a generic small constant
ϵ_t	Noise variable at step t (disambiguated by context)
$\mathcal{D}$	Experience replay buffer
$Z(s, a)$	Return distribution at (s, a)
δ	Temporal-difference error
K	Number of ensemble members
τ	Quantile fraction $\in [0, 1]$; or a temperature parameter
$\mathcal{T}_i$	i -th task in a meta-learning task distribution

- *Tabular Q-learning convergence and TD methods*: Sutton & Barto 2018, *Reinforcement Learning: An Introduction* (2nd ed.), chapters 6 and 7.
- *Function approximation and the deadly triad*: Sutton & Barto 2018, chapter 11; Tsitsiklis & Van Roy 1997.
- *Distributional RL theory*: Bellemare, Dabney & Rowland 2023, *Distributional Reinforcement Learning* (MIT Press).
- *Offline RL theory and pessimism-based methods*: Levine et al. 2020 survey; Jin, Yang & Wang 2021.
- *Multi-agent value decomposition*: Wang et al. 2020 (QPLEX); Albrecht & Stone 2018, *Multiagent Learning* (recent survey).

S6. TABULAR EXPERIMENT. CONFIGURATION AND REPRODUCTION

This section documents the full configuration behind the tabular study of §VI so that its numbers can be reproduced exactly. The experiment is implemented in a single self-contained NumPy/Gymnasium script (`tabular_experiments.py`, provided with this supplement); it requires no GPU, as tabular Q-learning is scalar table arithmetic with no tensor workload.

Protocol. For each environment and each model-free learner we train over 100 random seeds, then evaluate the greedy ($\arg\max_Q$) policy over 300 episodes per seed. We report the mean episodic return across seeds with a 95% bootstrap confidence interval (10,000 resamples of the per-seed means). Exploration uses an ϵ -greedy schedule decayed linearly from $\epsilon = 1.0$ to $\epsilon = 0.05$ over the first 80% of training and held at 0.05 thereafter. The n -step learner uses $n = 3$.

TABLE V

PER-ENVIRONMENT TRAINING CONFIGURATION. CLIFFWALKING HAS NO BUILT-IN EPISODE LIMIT, SO EPISODES ARE CAPPED AT 500 STEPS (TRUNCATION, WITH BOOTSTRAPPING CONTINUED) TO BOUND EVALUATION OF ANY NON-TERMINATING POLICY.

Environment	Episodes	α	γ	States $\times$ Actions
FrozenLake-v1 (4 $\times$ 4, slippery)	15,000	0.10	0.99	16 $\times$ 4
Taxi-v3	12,000	0.10	0.99	500 $\times$ 6
CliffWalking	2,500	0.50	0.99	48 $\times$ 4

Planning oracles. The Value/Policy Iteration upper bound is computed from the environment’s exact transition model P and reward R (exposed by Gymnasium for these tabular environments) by value iteration to a residual tolerance of 10^{-10} , with terminal-state values held at zero. Because these methods require full model access, they are reported only as a planning upper bound, never as a model-free competitor.

Full results. The table below reproduces §VI’s results with standard deviations across the 100 seeds added alongside the 95% bootstrap CIs.

Reproduction. Running the script with the default arguments (`--seeds 100`) regenerates the JSON results file that this table and §VI are built from; results are deterministic given the seed range. The run parallelizes trivially across CPU cores (one seed per worker).

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TABLE VI

FULL TABULAR RESULTS: MEAN $\pm$ STANDARD DEVIATION OVER 100 SEEDS, WITH 95% BOOTSTRAP CONFIDENCE INTERVALS IN BRACKETS. THE LARGE STANDARD DEVIATIONS FOR SARSA AND n -STEP Q ON CLIFFWALKING REFLECT GENUINE ACROSS-SEED INSTABILITY OF THEIR GREEDY POLICIES, NOT MEASUREMENT NOISE.

Algorithm	FrozenLake-v1 (success rate)	Taxi-v3 (return)	CliffWalking (return)
Q-learning	0.722 ± 0.090 [0.701, 0.737]	7.93 ± 0.16 [7.90, 7.96]	-13.0 ± 0.0 [-13, -13]
SARSA	0.692 ± 0.109 [0.669, 0.712]	7.90 ± 0.16 [7.87, 7.93]	-79.9 ± 162 [-114, -51]
Expected SARSA	0.720 ± 0.068 [0.705, 0.732]	7.93 ± 0.16 [7.90, 7.96]	-17.0 ± 0.0 [-17, -17]
n -step Q ($n=3$)	0.497 ± 0.228 [0.451, 0.541]	7.93 ± 0.16 [7.89, 7.96]	-75.2 ± 157 [-104, -46]
Planning oracle (VI/PI)	0.739	7.94	-13.0